單類別分類演算法

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OCC：單類別分類

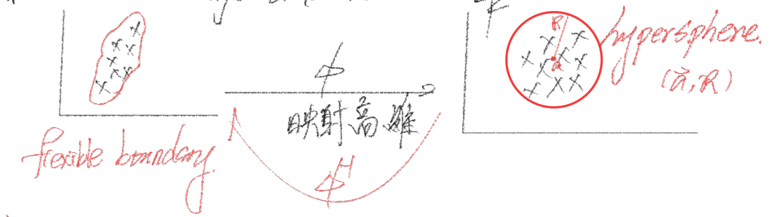
Given: data of one class (target class) $\vec{x}_i$ $i=1 \ldots N \in R^d$

OCC: to find a function to give a data domain description.

① density approach : GMM

② boundary approach : SVDD.

Supp: target data R^d



⇒ if x_i is test data, the SVDD decision boundfunction.

$$D(\vec{x}) = \| \phi(\vec{x}) - \vec{a} \|^2 - R^2$$

If $D(\vec{x}) \le 0$, $\vec{x}$ is accepted as a target data

else $D(\vec{x}) > 0$, $\vec{x}$ is rejected as an outlier

單類別分類 (one-class classification, OCC)

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- **OCC.** Most pattern recognition tasks deal with classification or regression problems. But there is a third, less well-known extension of the classification problem, the data domain description problem (also called one-class classification).

PR

描述 data point 在哪裡

- **Definition.** In domain description the task is **not** to distinguish between classes of objects like in classification problems or to produce a desired outcome for each input object like in regression problems, but to give a description of a set of objects. This description should cover the class of objects represented by the training set, and ideally should reject all other possible objects (data points) in the object space. ⇒ 須包蓋所有的 data, 並排除所有其它類型的資料

⇒ 並非是要區分資料類別

data

- **Application.** The data domain description is used for outlier detection or novelty detection, the detection of objects which differs in some sense significantly from the rest of the dataset.

離群值

異類

Why data domain description? S

Ex. $\vec{x}$ = unseen data

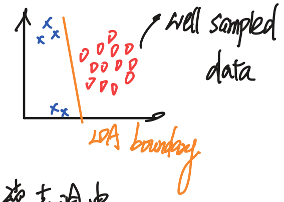
⇒ $\vec{x}$ belongs to S or not?

Why OCC? 為何要用 OCC

- Training data points for some classes (conditions) may not be available or representative.
- If the available training data of one class is under-sampled, the distribution of the training data cannot represent the class's true distribution. As a result, a binary classifier trained by such training data points (one class is well-sampled and the other is under-sampled) will give a "biased" decision boundary.
- OCC gives a new approach: since one class is under-sampled, OCC is trained only by the other class which is well-sampled.

Assume: motor → I: 电流訊号 → AI → (1) 正常 (2) 異常

Step 1. data collection: NG-data: $x_{N+1} \cdots x_M$: x
ok-data: $x_1 \cdots x_N$: D



⇒ Well Sampled Data: 可以充分代表這一類別資料, 在後續表現中

⇒ New NG data not included in the training set
也就是抽樣資料無法完全表示母體分布 (不具說服力)

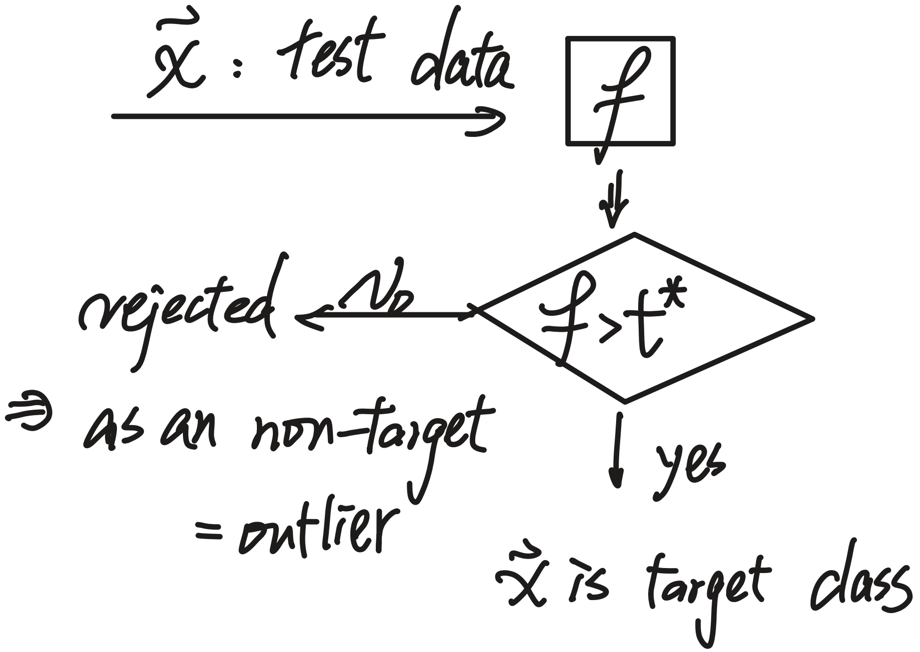
⇒ undersampled: 取樣不足 (抽樣分布不足以代表母体分佈)

OCC定義

- Well Sampled
  只用單一類別的資料訓練模型
- OCC definition : (手寫)

Assume: $\underbrace{\vec{x}_1 \cdots \vec{x}_N \in \mathbb{R}^d}_{\Rightarrow S}$ belong to the same class "Target"

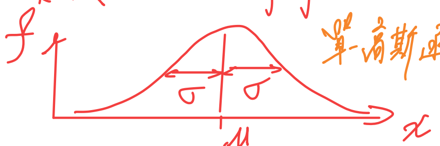
OCC: 用 S 建模 $f(x)$ 的函數



Ex: $\vec{x}_1 \dots \vec{x}_N \in$ target class

$\boxed{\text{class } f\text{: Gaussian} \quad f(x) = \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{(x-\mu)^2}{2\sigma^2}}}$

建模: 用 data 得到 σ, μ



單高斯函數

$\boxed{\text{test stage}}$ x: test data

if $f(x) > f^*$ (ex. 0.2)

then x is accepted as a target

else

x is rejected as an outlier

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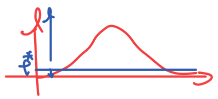
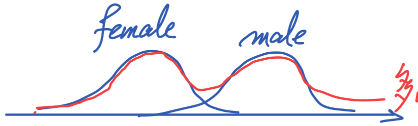
OCC方法 (1/2)

1. Density approach: Provide an estimate of the probability for a certain object given the model family. Low probabilities will then indicate a possible outlier.
   - Single Gaussian method 用 r.v. X 建立單一高斯模型
   - Bayesian approach 貝式定理: 條件機率
   - Gaussian mixture model (GMM)



$2 \le k \le n$ (data數) ⇒ 多模態分佈

X: r.v. : height



多峰分佈 (多模態分布) multi-modal distribution

GMM: f 由多個高斯函數組合

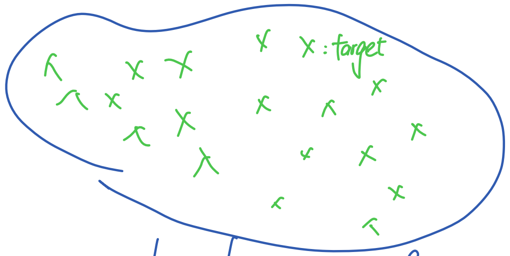
$$f(x) = w_1 f_1(x) + w_2 f_2(x) + \cdots + w_k f_k(x)$$

$$= w_1 \times \frac{1}{\sqrt{2\pi}\sigma_1} e^{\frac{-(x-\mu_1)^2}{2\sigma_1^2}} + w_2 \times \frac{1}{\sqrt{2\pi}\sigma_2} e^{\frac{-(x-\mu_2)^2}{2\sigma_2^2}} + \cdots + w_k \frac{1}{\sqrt{2\pi}\sigma_k} e^{\frac{-(x-\mu_k)^2}{2\sigma_k^2}}$$

OCC方法 (2/2)

2. Boundary approach : compute the boundary around a target data set
    - Support vector data description (SVDD)

3. Error reconstruction approach
    - Kernel PCA for novelty detection (2008)



boundary

f= D

(1) ad. (1) density independent

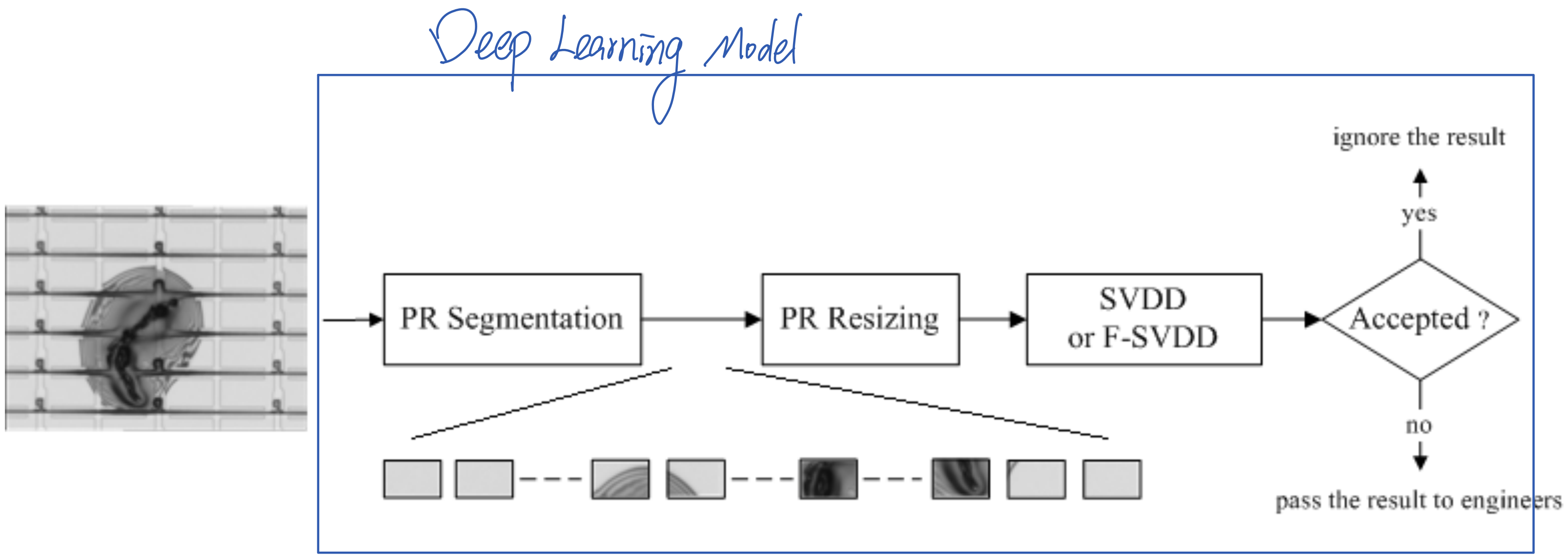
(2) N 不用大

(3) non linear distribution

⇒ 找出 non linear boundary

OCC Examples

- 馬達故障檢測
- 製程異常檢測
- 影像瑕疵檢測
- 生理訊號異常檢測



Liu et al., “Fast support vector data description for novelty detection,” *IEEE Trans. Neural Networks*, 2010.

協助教材內容編輯：

李威廷
曾昱嘉
官韋宏

Support Vector Data Description (SVDD)

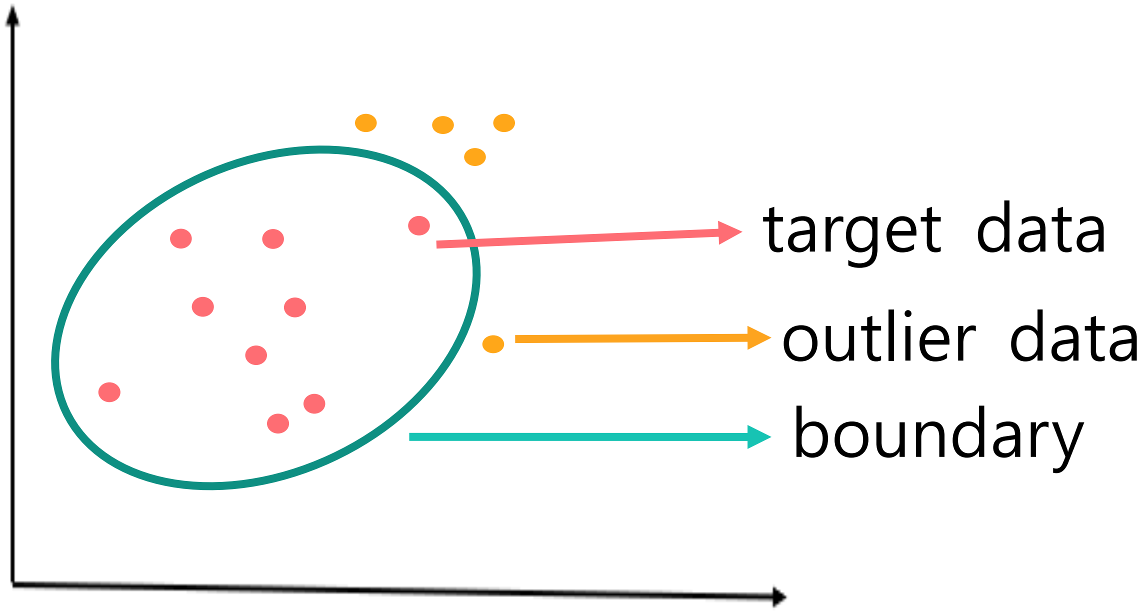
D. Tax and R. Duin, “Support vector data description,” *Machine Learning*, vol. 54, pp. 45–66, 2004.

SVDD-概念 (also called one-class SVM)

- Concept & idea：

Cover. 封閉的曲面

SVDD algorithm tries to find a boundary to enclose a set of target data.



判別 Rule.

If a test data **X** is inside of the boundary, then **X** is classified(accepted) as a target data； Else, it is outlier data.

SVM ⇒ SVDD

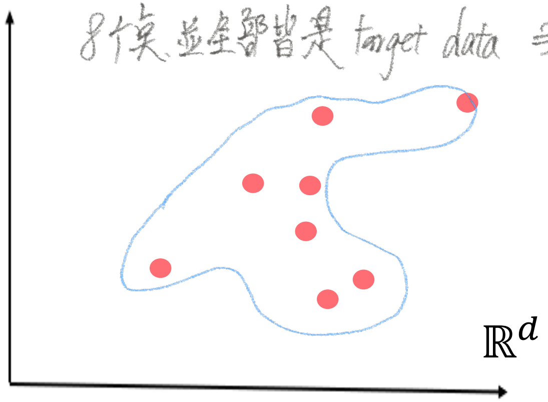
SVDD : two-step 概念

1. First, map the target data from $\mathbb{R}^d$ space into a higher-dimensional feature space **F** via a nonlinear-mapping：

映射至高維空間

習慣將球包緊
⇒避免出錯
∵雖會誤傷，但可補救

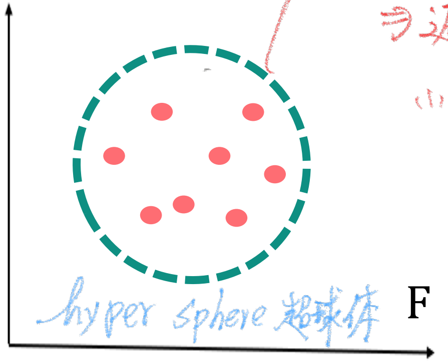
8個点並全部皆是 target data ⇒太散了 須使用不考慮density的方法



$$\phi: \ \mathbf{x}_i \in \mathbb{R}^d \rightarrow \phi(\mathbf{x}_i) \in \mathrm{F}$$

Nonlinear Mapping

ϕ



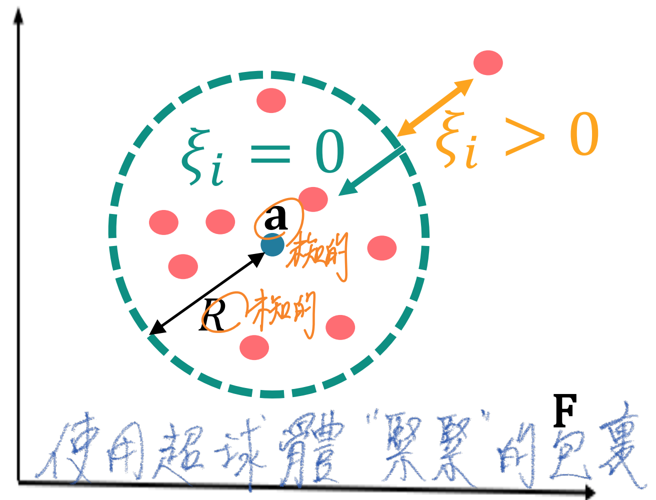
such that the mapped data $\phi(\mathbf{x}_i), i = 1, \ldots, N$ become a spherically-shaped distribution.

2. Then, try to find a hypersphere(超球體) to tightly enclose the mapped target training data in the space F .

SVDD- Formulation公式化

R ：Radius of the hypersphere
a ：Center of the hypersphere
c ：Penalty weight
ξ_i ：Slack variable
$\emptyset$ ：Nonlinear mapping

- SVDD欲找到最小體積的超球體，且要確保最多的mapped target data 在F 空間中。



使用超球體"聚聚"的包裹

希望找一個最小超球體，可以涵蓋越多點越好.

先前問題(Primal problem)

$$
\begin{cases}
\text{Minimize：} \quad R^2 + C\sum_{i=1}^{N}\xi_i \\
\text{subject to} \quad \|\emptyset(\mathbf{x}_i) - \mathbf{a}\|^2 \leq R^2 + \xi_i \,, \forall i = 1, \dots, N \\
\qquad\qquad\quad \xi_i \geq 0, \qquad \forall i = 1, \dots, N
\end{cases}
$$

(體積最小的球和包涵越多點)

平衡權重

半徑距離

⇒ 找出一組最好的 (a, R)，最小化 $R^2+C\Sigma\xi$

求解先前問題

先前問題(Primal problem)

$$\begin{cases} \underset{(R,\mathbf{a},\xi_i)}{\text{Minimize}}： & R^2 + C\sum_{i=1}^{N}\xi_i \\ \text{subject to}： & ❶\ ||\emptyset(\mathbf{x}_i) - \mathbf{a}||^2 \le R^2 + \xi_i,\ \forall i = 1,\dots,N \quad ❷\ \xi_i \ge 0, \forall i = 1,\dots,N \end{cases}$$

補充：Consider a constrained optimization problem

$\alpha,\ \beta$：Lagrange multipliers

$$\begin{cases} \text{Min/Max} & f(x) \\ \text{Subject to} & g(x) \ge 0 \\ & h(x) \ge 0 \end{cases} \longrightarrow$$

Lagrangian Function(L)：

$$\mathrm{L} \triangleq f(x) - \alpha g(x) - \beta h(x)$$

求解先前問題

Step 1：Construct the Lagrangian

$$\mathrm{L} = f(x) - \alpha g(x) - \beta h(x)$$

$$= R^2 + C\sum_{i=1}^{N}\xi_i - \sum_{i=1}^{N}\alpha_i\,[R^2 + \xi_i - ||\emptyset(\mathbf{x}_i) - \mathbf{a}||^2] - \sum_{i=1}^{N}\beta_i\,\xi_i$$

Step 2：Take partial differentiation w.r.t the primal variables R, $\mathbf{a}$, and ξ_i, and setting the results to be equal zero

i.e. $\frac{\partial \mathrm{L}}{\partial R} = 0$ $\quad \frac{\partial \mathrm{L}}{\partial \mathbf{a}} = 0$ $\quad \frac{\partial \mathrm{L}}{\partial \xi_i} = 0$

求解先前問題

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Step 2：

$$\mathrm{L} = R^2 + C\sum_{i=1}^{N}\xi_i - \sum_{i=1}^{N}\alpha_i\left[R^2 + \xi_i - ||\emptyset(\mathbf{x}_i) - \mathbf{a}||^2\right] - \sum_{i=1}^{N}\beta_i\,\xi_i$$

$$\frac{\partial \mathrm{L}}{\partial R} = 0 \rightarrow \cancel{2R} - \sum_{i=1}^{N}\alpha_i(\cancel{2R}) = 0 \rightarrow \sum_{i=1}^{N}\alpha_i = 1 \quad ❶$$

1 1

$$\frac{\partial \mathrm{L}}{\partial \mathbf{a}} = 0 \rightarrow -\sum_{i=1}^{N}\alpha_i\left[-\cancel{2}(\emptyset(\mathbf{x}_i) - \mathbf{a})\right] = 0 \rightarrow \sum_{i=1}^{N}\alpha_i\,\emptyset(\mathbf{x}_i) - \sum_{i=1}^{N}\alpha_i\,\mathbf{a} = 0$$

$$\rightarrow \mathbf{a} = \frac{\sum_{i=1}^{N}\alpha_i\,\emptyset(\mathbf{x}_i)}{\cancel{\sum_{i=1}^{N}\alpha_i}\ ❶} \rightarrow \mathbf{a} = \sum_{i=1}^{N}\alpha_i\,\emptyset(\mathbf{x}_i) \quad ❷$$

linear combination

$$\frac{\partial \mathrm{L}}{\partial \xi_i} = 0 \rightarrow C = \alpha_i + \beta_i \quad ❸$$

求解先前問題

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Step 3：Substitute 1 2 3 back into the Lagrangian function L

$$\mathrm{L} = \cancel{R^2} + \cancel{C\sum_{i=1}^{N}\xi_i} - \sum_{i=1}^{N}\alpha_i\left[\cancel{R^2} + \cancel{\xi_i} - ||\emptyset(\mathbf{x}_i) - \mathbf{a}||^2\right] - \cancel{\sum_{i=1}^{N}\beta_i\xi_i}$$

1 3

$$\rightarrow \sum_{i=1}^{N}\alpha_i\,||\emptyset(\mathbf{x}_i) - \mathbf{a}||^2$$

2

$$\rightarrow \sum_{i=1}^{N}\alpha_i\,\underbrace{\emptyset^{\mathrm{T}}(\mathbf{x}_i)\emptyset(\mathbf{x}_i)}_{k(x_i,x_i)} - \sum_{i=1}^{N}\sum_{j=1}^{N}\alpha_i\alpha_j\underbrace{\emptyset^{\mathrm{T}}(\mathbf{x}_i)\emptyset(\mathbf{x}_j)}_{k(x_i,x_j)}$$

可觀察到：
1. 所有的先前變數的不見
2. 只剩下變數 $\alpha_1 \ldots \alpha_N$ 要求解

求解先前問題

得到Lagrange multipliers($\alpha_i \ldots \alpha_N$)來解決對偶問題(Dual problem)，如下列

對偶問題(Dual problem)

SVDD

$$\begin{cases} \underset{(\alpha_1 \ldots \alpha_N)}{\text{maximize}} & \sum_{i=1}^{N} \alpha_i \emptyset^{\mathrm{T}}(\mathbf{x}_i)\emptyset(\mathbf{x}_i) - \sum_{i=1}^{N}\sum_{j=1}^{N} \alpha_i\alpha_j \emptyset^{\mathrm{T}}(\mathbf{x}_i)\emptyset(\mathbf{x}_j) \\ \text{subject to} & \sum_{i=1}^{N} \alpha_i = 1 \quad \text{equality constraint} \end{cases}$$

$$\Rightarrow \sum_{i=1}^{N} \alpha_i k(x_i, x_i) - \sum_i \sum_j \alpha_i \alpha_j k(x_i, x_j)$$

$$0 \leq \alpha_i \leq C, \forall i = 1, \ldots, N$$

$\Rightarrow$ 找出 $\alpha_1 \ldots \alpha_N$ 的 optimal solution set

optimal solution set

求解先前問題

若不知$\phi(x_i)$則不知球型 range
$\Rightarrow$ 解出dual prob.才知α
$\Rightarrow$ 須先確認 kernal function 為何

- $\emptyset^{T}(\mathbf{x}_i)\emptyset(\mathbf{x}_i)$藉由kernal的概念可以表示成$K(\mathbf{x}_i, \mathbf{x}_i)$

對偶問題(Dual problem)

$e^{-\frac{0}{2\sigma^2}} = 1$

$$\begin{cases} \text{Maximize}_{(\alpha_1 \dots \alpha_N)} & \boxed{\sum_{i=1}^{N} \alpha_i}\, \mathrm{k}(\mathbf{x}_i, \mathbf{x}_i) - \sum_{i=1}^{N}\sum_{j=1}^{N} \alpha_i \alpha_j K(\mathbf{x}_i, \mathbf{x}_j) \\ \text{subject to} & ① \sum_{i=1}^{N} \alpha_i = 1 \quad \text{equality constraint} \quad ② \; 0 \le \alpha_i \le C, \forall i = 1, \dots, N \end{cases}$$

$=1$

Note：

If the kernel is a Gaussian Kernel：$K(\mathbf{x}_i, \mathbf{x}_i) = \emptyset^{T}(\mathbf{x}_i)\emptyset(\mathbf{x}_i) = e^{-\frac{\|\mathbf{x}_i - \mathbf{x}_i\|^2}{2\sigma^2}} = 1$

Then the objective function is： Maximize $1 - \sum_{i=1}^{N}\sum_{j=1}^{N} \alpha_i \alpha_j K(\mathbf{x}_i, \mathbf{x}_j)$

Kuhn-Tucker(KT) condition 求R值

定義：若problem為
$$\begin{cases} \text{Min/Max} & f(x) \\ \text{subject to} & g(x) \geq 0 \\ & h(x) \geq 0 \end{cases}$$

則其KT condition 為
$$\begin{cases} \alpha g(x) = 0 \\ \beta h(x) = 0 \end{cases}$$

一次只能其中一個為0
例如：$\alpha g(x) = 0 \rightarrow \alpha = 0$ or $g(x) = 0$

i.e. 若$\alpha g(x) = 0$；若$\alpha = 0$ 則 $g(x) > 0$；若$g(x) = 0$ 則 $\alpha > 0$

There are 3 possible cases of α_i：

① $\alpha_i = 0$
② $\alpha_i = C$
③ $0 < \alpha_i < C$

利用Kuhn-Tucker(KT) condition 求R值(Case 1)

KT conditions of SVDD：$\begin{cases} \alpha_i[R^2 + \xi_i - ||\emptyset(\mathbf{x}_i) - \mathbf{a}||^2] = 0 \\ (C - \alpha_i)\,\xi_i = 0 \quad , \forall i = 1, \dots, N \end{cases}$

$\because \alpha_i = 0$ 且 $|| \quad || > 0$

$\Rightarrow R^2 + \cancel{\xi_i}^{\,0} - ||\phi(x_i) - \vec{a}||^2 > 0$

$\Rightarrow \underbrace{R^2 - ||\phi(x_i) - \vec{a}||^2}_{\text{與球心距離}} > 0$

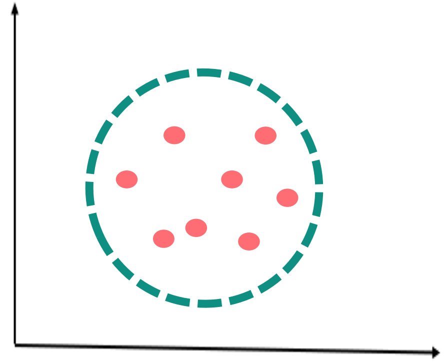
$\beta_i \xi_i = 0$

Case 1：若$\alpha_i = 0 \rightarrow C > 0 \rightarrow \xi_i = 0$

$\Rightarrow C - \alpha_i = \beta_i > 0 \Rightarrow \xi_i = 0$ (圓內)

$$\rightarrow \quad R^2 - ||\emptyset(\mathbf{x}_i) - \mathbf{a}||^2 > 0$$

$$\rightarrow \quad R > ||\emptyset(\mathbf{x}_i) - \mathbf{a}||^2$$



表示$\emptyset(\mathbf{x}_i)$在超球體裡面

利用Kuhn-Tucker (KT) condition 求R值(Case 2)

Case 2：$0 < \alpha_i < C$　　∵ $C - \alpha_i > 0$　∴ $\xi_i = 0$

$\alpha_i \neq 0 \Rightarrow C - \alpha_i > 0$

$\Rightarrow \beta_i > 0 \Rightarrow \xi_i = 0$ (球面上))

$\underbrace{\alpha_i}_{\neq 0}[R^2 + \cancel{\xi} - \|\phi(\vec{x}_i) - \vec{a}\|^2] = 0$

$\Rightarrow R^2 - \|\phi(x_i) - a\|^2 = 0$

$\Rightarrow \|\phi(x_i) - a\|^2 = R$ (球上)

$$\rightarrow \quad R^2 - ||\emptyset(\mathbf{x}_i) - \mathbf{a}||^2 = 0$$

$$\rightarrow \quad R^2 = ||\emptyset(\mathbf{x}_i) - \mathbf{a}||^2$$

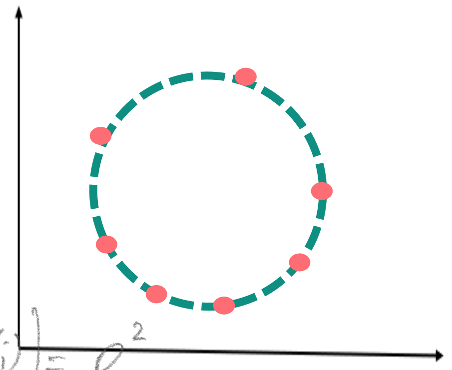
$\vec{x}_k =$ $0 < \alpha_k < C$

$\Rightarrow \|\phi(\vec{x}_k) - a\|^2 = R^2$

$\Rightarrow \phi(\vec{x}_k)^T\phi(\vec{x}_k) - 2\vec{a}^T\phi(x_k) + \vec{a}^T a = R^2$

$\Rightarrow \phi(\vec{x}_k)^T\phi(\vec{x}_k) - 2\sum_{i=1}^{N}\alpha_i\phi(\vec{x}_i)^T\phi(\vec{x}_k) + \left(\sum_{i=1}^{N}\alpha_i\phi(\vec{x}_i)\right)^T\left(\sum_{j=1}^{N}\alpha_j\phi(\vec{x}_j)\right) = R^2$

$R = \sqrt{K(\vec{x}_k,\vec{x}_k) - 2\sum_{i=1}^{N}\alpha_i K(\vec{x}_i,\vec{x}_k) + \sum_{i=1}^{N}\sum_{j=1}^{N}\alpha_i\alpha_j K(\vec{x}_i,\vec{x}_j)}$



表示$\emptyset(\mathbf{x}_i)$在超球體表面

$0 \leq \alpha_i \leq C$

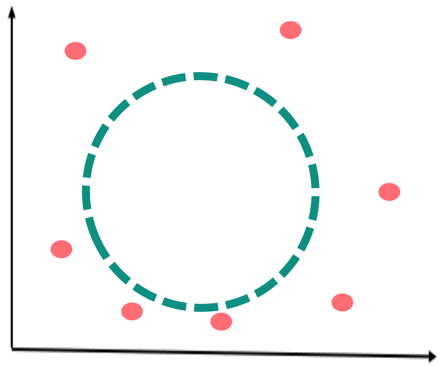
利用Kuhn-Tucker (KT) condition 求 R值(Case 3)

Case 3：$\alpha_i = C$ $\quad \because C - \alpha_i = 0 \;\therefore \xi_i > 0$

$\beta_i = C - \alpha_i = 0 \quad \xi_i \neq 0$

$\rightarrow R^2 + \xi_i - ||\emptyset(\mathbf{x}_i) - \mathbf{a}||^2 = 0$

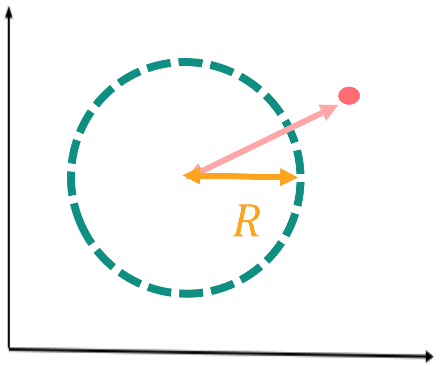
$\rightarrow R^2 = ||\emptyset(\mathbf{x}_i) - \mathbf{a}||^2 - \xi_i$



表示$\emptyset(\mathbf{x}_i)$在超球體外面

SVDD's Decision function

x : test data



$$\mathrm{D}(\mathbf{x}) = \left(\begin{array}{l}\text{The distance between } \emptyset(\mathbf{x}) \\ \text{and the center of the hypersphere}\end{array}\right) - \left(\text{Radius of the hypersphere}\right)$$

$$= ||\emptyset(\mathbf{x}) - \mathbf{a}||^2 - R^2$$

$$= ||\emptyset(\mathbf{x})||^2 - 2\emptyset^{\mathrm{T}}(\mathbf{x})\mathbf{a} + \mathbf{a}^{\mathrm{T}}\mathbf{a} - R^2$$

$$= \emptyset^{\mathrm{T}}(\mathbf{x})\emptyset(\mathbf{x}) - 2\emptyset^{\mathrm{T}}(\mathbf{x}_i)\sum_{i=1}^{N}\alpha_i\,\emptyset(\mathbf{x}_i) + \left(\sum_{i=1}^{N}\alpha_i\,\emptyset(\mathbf{x}_i)\right)^{\mathbf{T}}\left(\sum_{j=1}^{N}\alpha_j\,\emptyset(\mathbf{x}_j)\right) - R^2$$

未知

訓練資料筆數

$$\boxed{\mathrm{D}(\mathbf{x}) = \mathrm{k}(\mathbf{x},\mathbf{x}) - 2\sum_{i=1}^{N}\alpha_i\,\mathrm{k}(\mathbf{x},\mathbf{x}_i) + \sum_{i=1}^{N}\sum_{j=1}^{N}\alpha_i\alpha_j\mathrm{k}(\mathbf{x}_i,\mathbf{x}_j) - R^2}$$ $\propto O(N)$

training data　　半徑

大部分 α_i 為 0.　　已知 Const.　　已知 Const.

Decision function

⇒ 只須要對 support vector. α_i 作計算

$D(\vec{x}) = K(x,x) - 2\sum_{x_i \in SV} \alpha_i K(x_i,x) + \text{Const.}$

※ 若 kernal 是 Gaussian

$\Rightarrow K(x,x) = 1$

$D(\vec{x}) = 1 - 2\sum_{x_i \in SV} \alpha_i K(x_i,x) + \text{const} \quad \propto \quad D(\underline{\#SV})$

number of support vector

If $\mathrm{D}(\mathbf{x}) > 0$, $\mathbf{x}$ is rejected as an Outlier.

$\mathrm{D}(\mathbf{x}) < 0$, $\mathbf{x}$ is accepted as a Target data point.

Def：

The training data point $\mathbf{x}_i$ is a Support Vector if $\alpha_i \neq 0$.

性能指標 Performance index for OCC.

1. Target acceptance rate (TAR) = $\frac{\text{\# accepted data}}{\text{\# target data}}$ target data accepted : 越高越好.
   目標接受率
2. Target rejection rate (TRR) = $\frac{\text{\# target data rejected}}{\text{\# total target data}}$ 越低越好
3. Outlier rejection rate (ORR) = $\frac{\text{\# outlier rejected}}{\text{\# total outlier}}$ 越高越好. = 1-OAR
4. Outlier acceptance rate (OAR) = $\frac{\text{\# outlier accepted}}{\text{\# total outliers}}$ 越少越好
   error
5. 分類率 (classification rate) = $\frac{\text{\# target accepted + outlier rejected}}{\text{\# total data (outlier + target)}}$ 易有偏差.
6. 平衡損失 (balanced loss, BL) = $\frac{1}{2}(TRR+OAR)$
   error.
   $\Rightarrow TRR + OAR$

相互關聯:

TAR=1-TRR

SVDD的超參數調整：C and kernel parameter (1/2)

min $R^2+C\sum\xi$

- 懲罰權重 C 的可調整範圍如下:

研究上C不能亂設定

$\sum \alpha_i = 1$

$0 \le \alpha_i \le C$

$\sum_i^N \alpha_i \le \sum_i^N C$

$1 \le NC$

$\frac{1}{N} \le C \le 1$

$\sum_i \alpha_i = 1 \Rightarrow \alpha_i < C$

$\Rightarrow \sum_c C < 1$

$\Rightarrow C < 1$

$\sum C = 1$ 即每個 α_i 用C代，有N個C

$\sum_{i=1}^N \alpha_i = 1$ 不能大於1

$\alpha_i < C$

$\sum C = 1$

$C < 1$

0, 0, 0, 0 ... 0 C=1

- 懲罰權重 C 值大小對SVDD的影響

若C↑，$\sum_i \xi_i$ term 在訓練中越重要，傾向減少 $\sum \xi_i$

⇒ $\sum \xi_i$ 越小

⇒ TAR ↑ ⇒ 接收 range ↑ 球越大 ⇒ boundary 越鬆

⇒ DRR ↓ 可能下降

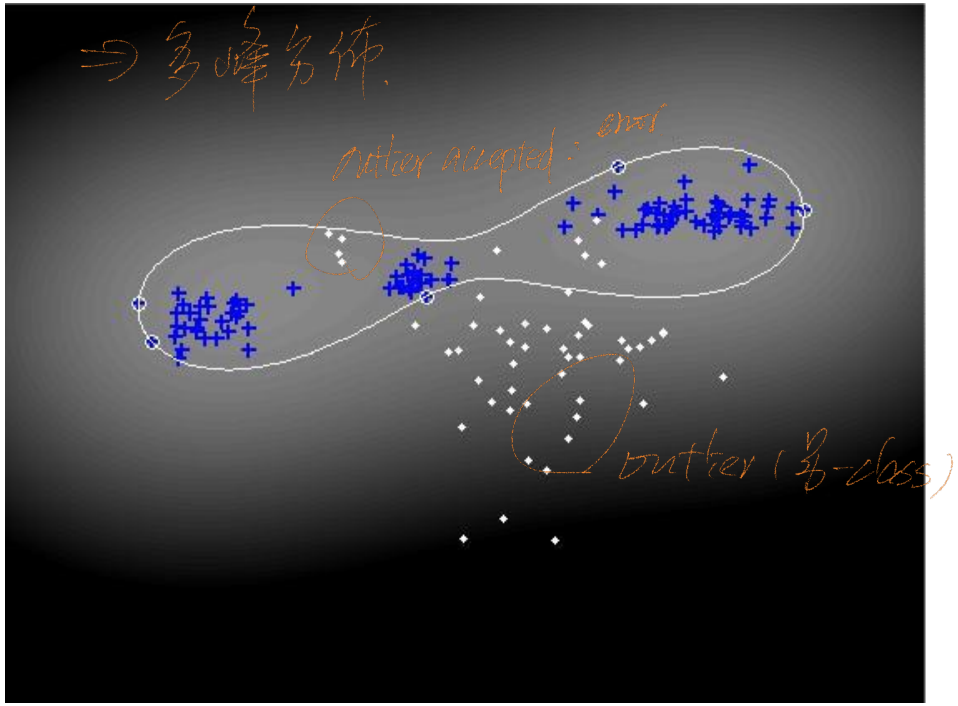
* 包的 boundary 是弄包的越緊越好

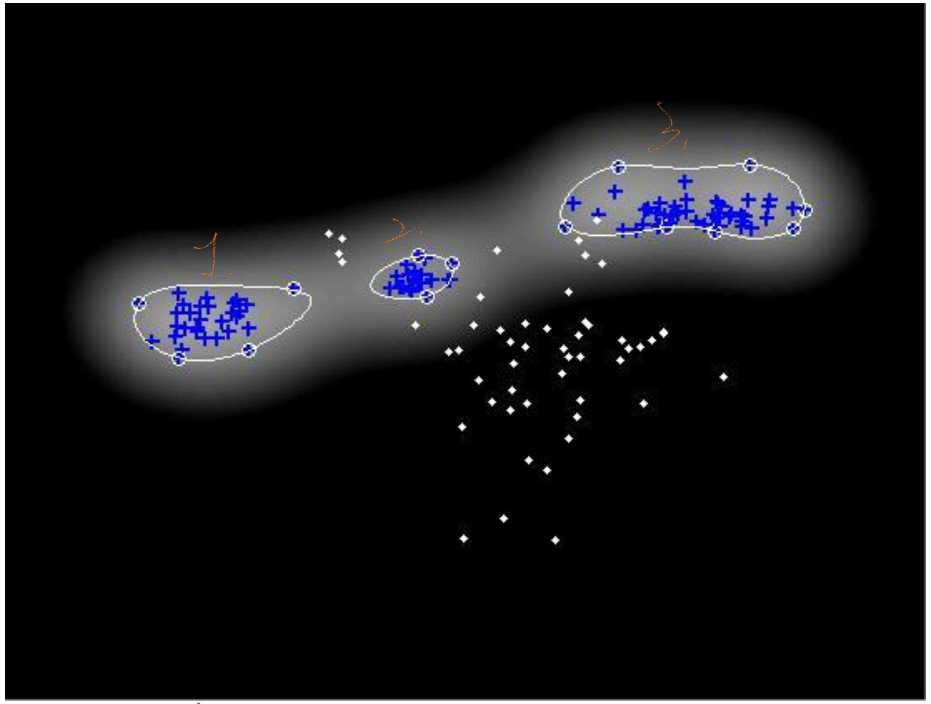
寧可錯殺不能放過一個

⇒ C ↓

SVDD的超參數調整：C and kernel parameter (2/2)

- 以Gaussian kernel 為例(kernel parameter is σ)說明如下：





σ value is larger — σ 調小, 以緊縮 SVDD boundary → *σ value is smaller*

⇒ σ 越大, boundary 越鬆.

* 當σ→∞ = 在原空間中找超球体.

SVDD的超參數調整：C and kernel parameter (2/2)

Idea: 只有 target data 時，如何 train 合適的 OCC model
(合理)

Why don't use optimal OCC model ?

⇒ 只需能達到夠高的 TAR 即可 (ex. 95%, 97%)

(因 100% 可能造成 boundary 太鬆)

* 去頭去尾的 2 倍標準差，可以有效包含 96% 的可能值)

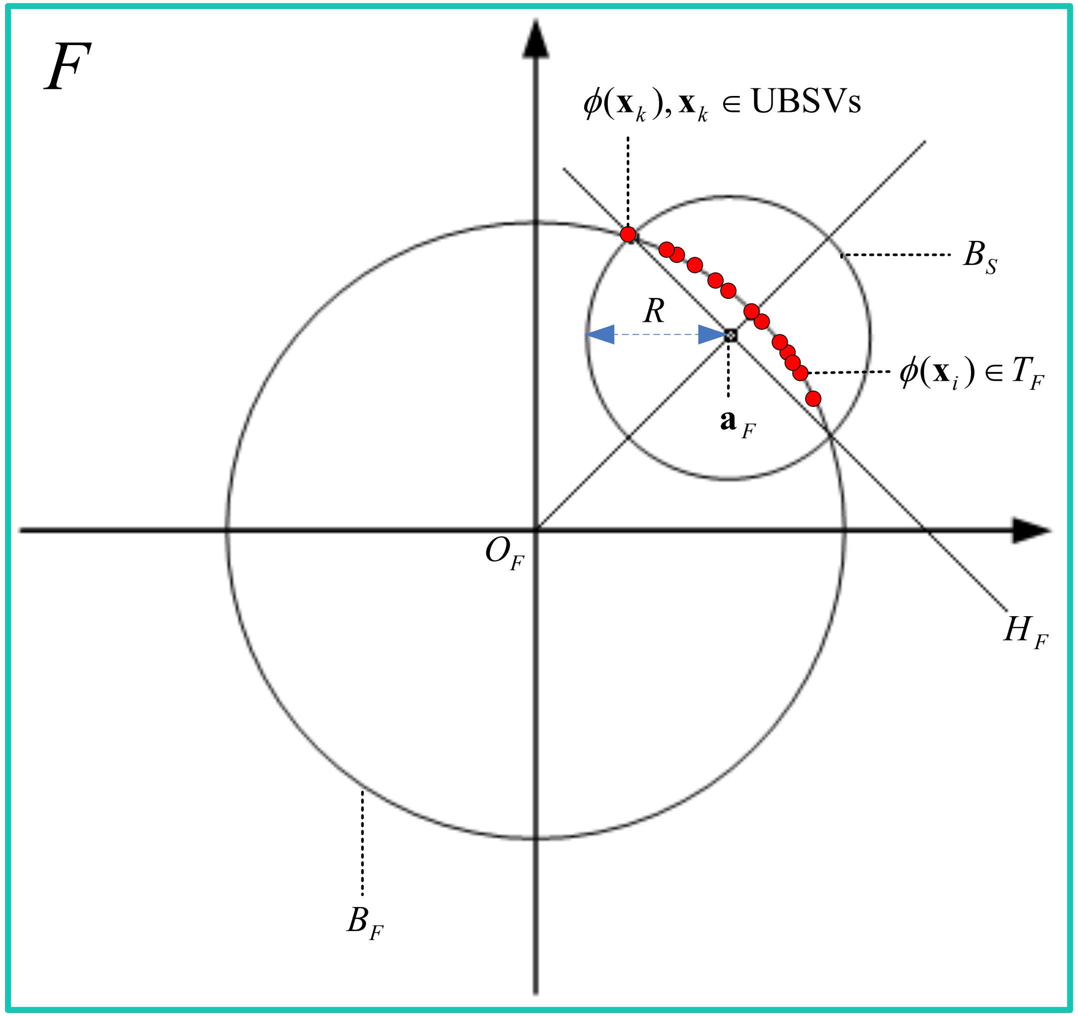
Ex. 調整 C, σ, such that TAR achieve 97%

SVDD training strategy (第一種方法)

- Given a target training set T $= \{x_i\}_{i=1 \sim N}$
- Choose a kernel (if a RBF kernel is chosen)
- Set a desired target rejection rate ($t^* = 3\%$ for example) fix $C=0.5$ TAR=97%
- Initialize the penalty weight C and the kernel parameter σ ⇒ 初始σ給大一些
- Training :
  1. Step 1. Train an SVDD with the set T. If the target rejection rate on the same training set T is lower than t^*, go to step 2.
  2. Step 2. Adjust the value of the kernel parameter σ . Repeat Step 1 until the actual target rejection rate is the same (or almost the same) as the predefined target rejection rate t^*. 調小 $\sigma = \sigma - \Delta\sigma$

for all OCC model
都可用

Geometric property of the SVDD hypersphere



Geometrical relationship between the SVDD sphere and the unit ball in the feature space F induced by a Gaussian kernel.